

Xavier Lermusieaux

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EDUCATION

University of Ottawa

Bachelor of Applied Science (BASc) in Software Engineering, Co-op

Ottawa, ON

Expected December 2026

- **CGPA:** 3.7/4.0 | **Honours & Awards:** Dean's Honour List; WaffleHacks 2023 Winner

EXPERIENCE

Ross Video

Software Developer, Robotics (Co-op)

Ottawa, ON

May 2026–August 2026

- Cut deployment setup by **6+ hours** by replacing manual XML layout edits with a React, TypeScript, and PixiJS 2D editor supporting translation, resizing, rotation, vertex editing, grid snapping, copy/paste, and robot drag-and-drop.
- Kept collaborative edits **responsive without sacrificing server consistency** by implementing optimistic WebSocket JSON Patch updates with automatic rollback, session takeover, and editor error recovery.
- Drove the **full-stack migration of Studio editing** from legacy REST screens to a WebSocket resource platform, unifying 5 workflows under multi-user drafts, validation, publishing, and MariaDB persistence.

Software Developer, Robotics (Co-op)

May 2025–August 2025

- Made schema migrations and type-safe SQL generation **reproducible against the production MariaDB engine**, replacing Liquibase with versioned Flyway scripts and packaging containerized setup in a reusable Gradle plugin.
- Improved configuration reliability across **4 product domains**—robots, routers, stations, and services—by adding server-side validation, field-level error reporting, and integration tests for duplicate IPs and missing data.
- Delivered full-stack Studio and Show management for robot assignment, ordering, and visual identity, adding live cross-session updates to **prevent stale views during concurrent configuration**.

Innovation, Science and Economic Development Canada (ISED)

Software and Operations Officer (Co-op)

Ottawa, ON

September 2024–December 2024; May 2023–August 2023

- Cut manual QA effort by **40%** while raising Salesforce regression test coverage to **85%** with Apex unit tests.
- Saved **2–3 hours of manual work** and reduced user-error risk by automating repeatable data validation and reporting workflows with VBA and SQL across multiple departments.
- Reached **up to 5,000 engineers** with program updates by building an automated Salesforce notification workflow.

PROJECTS

BinGo | Go, TypeScript, Systems Programming, DAP, WebSockets

2026

- Built a **cross-platform visual concurrency debugger** that makes Go runtime behavior observable through execution control and live views of goroutine lifecycles, stacks, variables, and source locations in VS Code.
- Implemented **2 native debugger backends** for Linux/amd64 and macOS/arm64 using ptrace, Mach exception ports, architecture-specific breakpoints, and DWARF data, enabling stepping, stack unwinding, and runtime inspection.
- Unified the debugger engine, session-based WebSocket server, and DAP adapter around shared sessions, allowing IDE, terminal, and automation clients to **control and observe the same program concurrently**.

QuickCV | Python, PyTorch, YOLOv8, OpenVINO, ONNX, OpenCV

2026

- Fine-tuned YOLOv8 on a 4-class KITTI-derived dataset (7,481 images; 35,950 objects), achieving **0.946 mAP@0.5** on 1,497 validation images, including 0.983 Car and 0.982 Truck AP.
- Engineered a confidence-gated YOLOv8n-to-YOLOv8m cascade with class-aware relabeling, escalating **32.1%** of INT8 detections and averaging a **+0.27 confidence gain** across 108 stage-two crop detections.
- Implemented ONNX/OpenVINO export and CPU benchmarking; INT8 cut **combined model size 74.0%** (116.0 to 30.2 MB), **wall time 47.5%** (22.08 to 11.59 s/100 images), and **stage-two latency 57.0%** (98.36 to 42.28 ms/crop).

TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, Go, C++, C, SQL | **Spoken:** Native English and French

Core: Data Structures & Algorithms, Object-Oriented Design, Concurrency, Debugging, Systems Programming

Frameworks & Libraries: React, Spring Boot, PyTorch

Databases & APIs: PostgreSQL, MariaDB, REST APIs, WebSockets, OpenAPI

Testing: Unit Testing, Integration Testing, JUnit, pytest, Testcontainers

Tools & Platforms: Git, Linux, Docker, Gradle, CMake, Flyway, GitHub Actions, Jenkins, CI/CD